

Mitochondrial iron chelation ameliorates cigarette-smoke induced bronchitis and emphysema in mice

SCIFACT-EVIDENCE-3553087

Evidence abstract

Chronic obstructive pulmonary disease (COPD) is linked to both cigarette smoking and genetic determinants.

We have previously identified iron-responsive element-binding protein 2 (IRP2) as an important COPD susceptibility gene and have shown that IRP2 protein is increased in the lungs of individuals with COPD.

Here we demonstrate that mice deficient in *Irp2* were protected from cigarette smoke (CS)-induced experimental COPD.

By integrating RNA immunoprecipitation followed by sequencing (RIP-seq), RNA sequencing (RNA-seq), and gene expression and functional enrichment clustering analysis, we identified *Irp2* as a regulator of mitochondrial function in the lungs of mice.

Irp2 increased mitochondrial iron loading and levels of cytochrome c oxidase (COX), which led to mitochondrial dysfunction and subsequent experimental COPD.

Frataxin-deficient mice, which had higher mitochondrial iron loading, showed impaired airway mucociliary clearance (MCC) and higher pulmonary inflammation at baseline, whereas mice deficient in the synthesis of cytochrome c oxidase, which have reduced COX, were protected from CS-induced pulmonary inflammation and impairment of MCC.

Mice treated with a mitochondrial iron chelator or mice fed a low-iron diet were protected from CS-induced COPD.

Mitochondrial iron chelation also alleviated CS-induced impairment of MCC, CS-induced pulmonary inflammation and CS-associated lung injury in mice with established COPD, suggesting a critical functional role and potential therapeutic intervention for the mitochondrial-iron axis in COPD.

Document metadata

Corpus | SciFact

Document ID | 3553087